



Technical Service Bulletin

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Phase 2.0 Harness Introduction

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This Technical Service Bulletin supersedes 04T19-2. Replace 04T19-2 with this topic.

This Technical Service Bulletin introduces the new QSK60 Industrial engine harness and the CENSE™ harness, also known as the Phase 2.0 harness. The new harness supersedes the Phase 1.0 harness in production, but the Phase 1.0 harness will be available for service. Installation procedures for the harnesses will be published in a separate publication. The engine serial number (ESN) first for this change is 33152044.

The new harness is a foam-filled metallic enclosure and designed to increase certain sensor, connector, and wiring robustness. New wiring diagrams will be available for the harness. The engine harness consists of two parts: one with three metallic enclosures linked together with two flexible conduits mounted on the left bank; the other is a single metallic enclosure mounted at the rear. The optional CENSE™ harness is mounted on the right bank.

Do **not** drill into metallic harness enclosures.

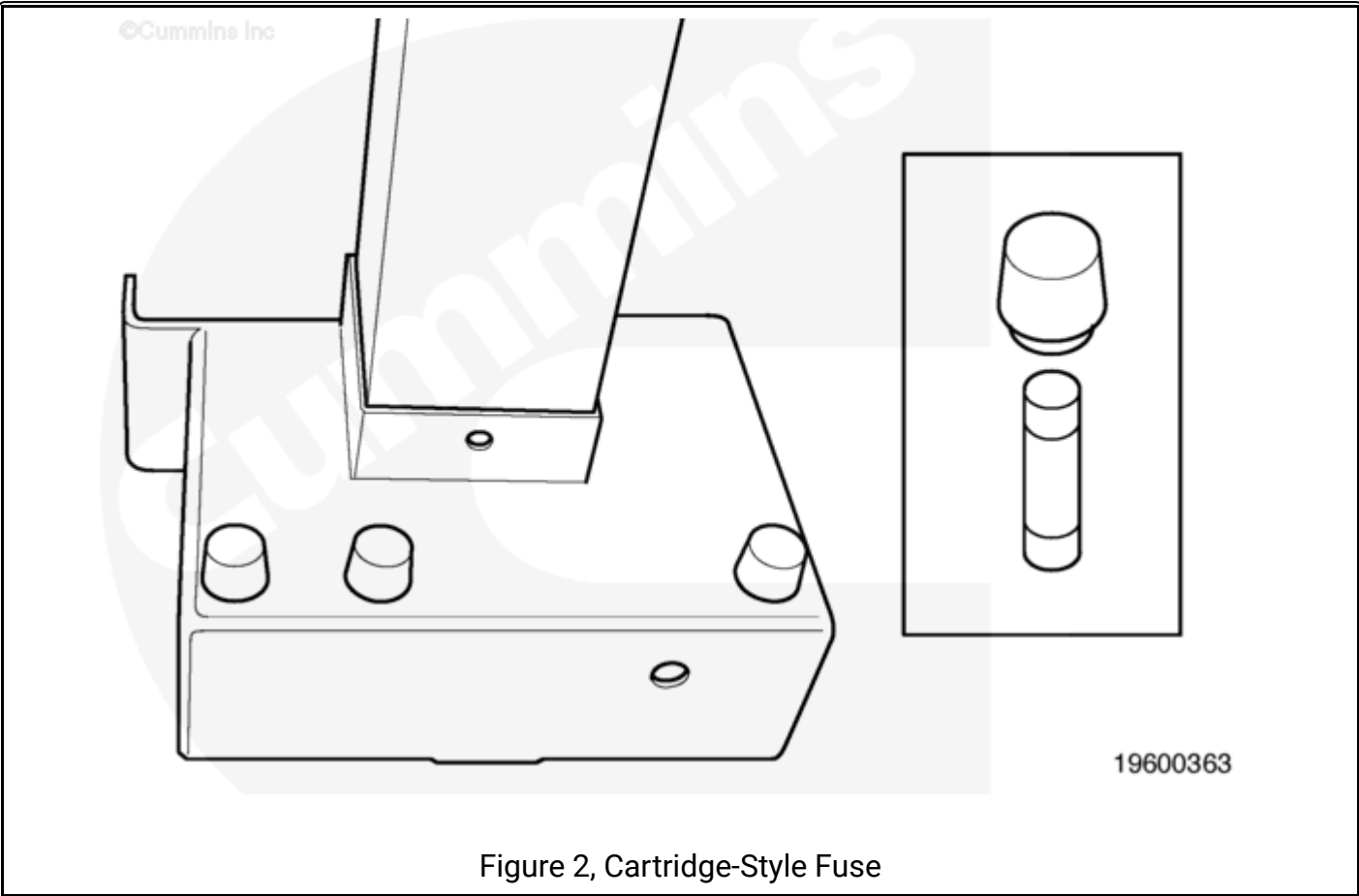
Engine Harness Overview

- The service port box section is mounted on the left bank front intake manifold.
- The rear section is connected to the left and right bank sections via 33- and 47-pin Deutsch connectors.
- Certain pressure sensors are being replaced with a more robust design.

The lower front and middle sections are mounted on the left bank cam follower covers while the third upper front section is mounted on the left bank front intake manifold. The separate rear section is mounted above the flywheel housing. A special bracket is used for engines with the '0' flywheel housing option. The locations for the mounting capscrews for the front and middle harness enclosures are shown in the installation procedure.

The original Phase 2.0 harness was designed with cartridge-style fuses. The new improved fuse is a Packard-style design. There are different part numbers for the left bank harness options (with CENSE™ and without CENSE™) for the different fuse designs.

Harness	FuseStyle	Part Number
LB harness with CENSE™	Cartridge Packard	4066648
		4101013
LB harness with CENSE™	Cartridge Packard	4066649
		4101012



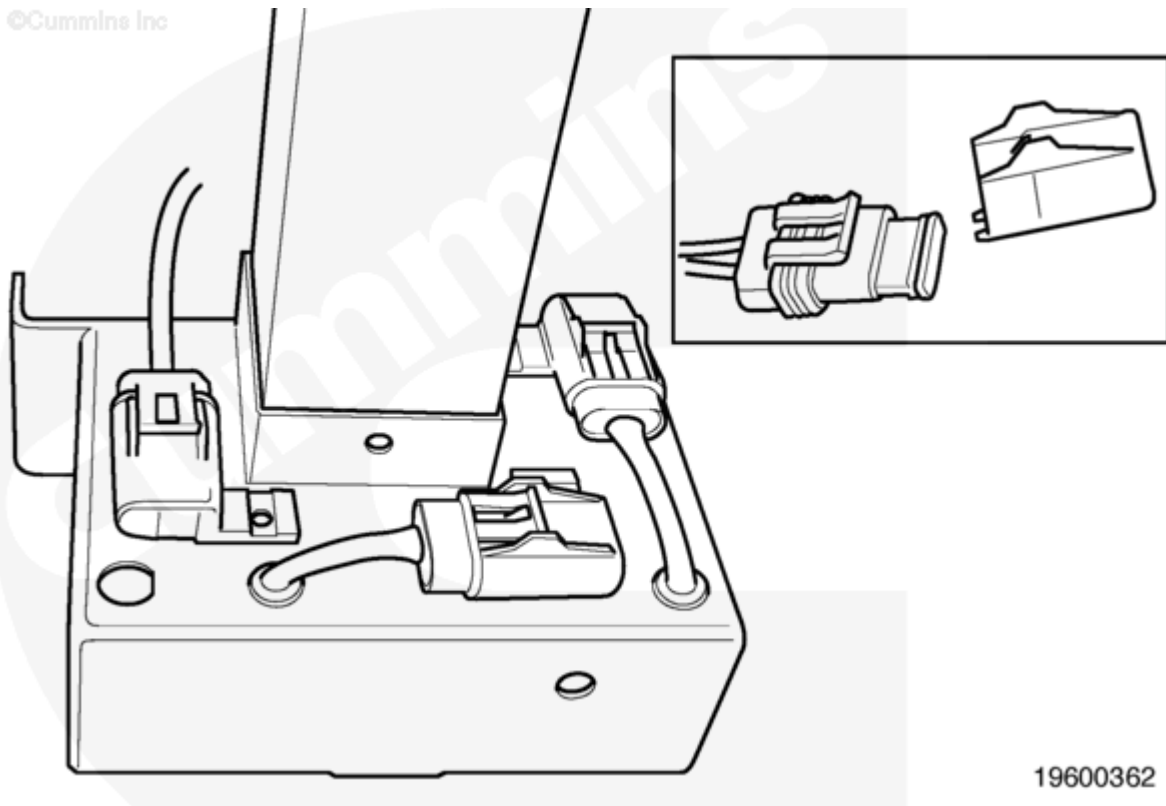


Figure 1, Packard-Style Fuse

A new version of the QSK60 wiring harness (Phase 2.0 harness) has been developed to improve product quality. The new features include:

- Enclosed aluminum harness pipe
- Danfoss brand pressure sensors
- DIN-style connectors
- Improved exhaust gas temperature sensor thermocouple geometry.

These changes increase sensor and wiring harness robustness. The Phase 2.0 harness specifically improves the reliability and durability of the CENSE™ module and the engine wiring harness.

The Phase 2.0 harness went into production November 25, 2002. The engine serial number first is 33153219.

Phase 2.0 electronics are a major improvement in the design and installation of sensors and Quantum and CENSE™ harnesses.

Key Design Features

Engine Harness

- Phase 2.0 harness employs foamed-pipe technology where the wires are enclosed inside a metallic box section that is foam filled and completely sealed.
- The Quantum harness is mounted on the left bank of the engine.
- There are four metallic enclosures linked together with flexible conduits.

- Two sections are mounted on the left bank cam follower covers.
- Service port box section is mounted on the left bank front intake manifold.
- Rear section is mounted on a bracket that is fitted to the flywheel housing.
- The rear section is connected to the left and right bank sections via 40-pin Deutsch connectors.

CENSE™ Harness

- CENSE™ electronic control module is mounted on the left bank number 3 cam follower cover
- CENSE™ harness is divided into two sections. For the Right Bank sensors/connections, there is a harness box section mounted on the right bank. It is linked to the CENSE™ electronic module on the left bank via a 40-pin Deutsch connector mounted on the rear harness section.
- The left bank connections are built into the left bank harness.
- The connections for the CENSE™ electronic module are incorporated into the left bank harness.

Pressure Sensors and Connection (Danfoss)

- The pressure sensor design is very robust.
- The DIN-style connector for these sensors has a locking feature that guarantees a firm connection between the sensor and harness. This results in improved robustness in terms of pin fretting.
- There is no need for a calibration change as a result of the revision to Danfoss sensors.
- These connectors are also keyed and color coded to prevent wrong connections being made in the field as well as on the assembly line.
- There is more information concerning the Danfoss sensors in the QSK45 Phase 2.0 harness introduction topic 03T19-30.

Pressure Sensor Part Number Service Adapter and Pencil Connector Matrix							
Pressure Sensor	DIN-Packard Connect or Adapter Harness	Old Sensor Part Number	New Sensor Part Number	DIN Connect or Keying	DIN Connect or Color	Packard Connect or Keying	Service Pigtail Part Number
Boost (left bank front)	3408507	3408347	3408556	Key 3	Green	Round	3164710
Boost (right bank rear)	3408507	3408347	3408556	Key 3	Green	Round	3164710
Coolant	3408508	3160404	3408556	Key 3	Green	Chevy	3164710
Fuel pump	3408509	3408414	3408566	Key 1	Black	Round	3164709

Pressure Sensor Part Number Service Adapter and Pencil Connector Matrix							
Pressure Sensor	DIN-Packard Connect or Adapter Harness	Old Sensor Part Number	New Sensor Part Number	DIN Connect or Keying	DIN Connect or Color	Packard Connect or Keying	Service Pigtail Part Number
Fuel rail	3408510	3408379	3408562	Key 2	Gray	Barn	3164709
Oil rifle	3408511	3865337	4088734	Key 2	Gray	Chevy	3164709
Post-filter oil	3408509	3330527	3408558	Key 1	Black	Round	3164708
Pre-filter oil	3408509	3330527	3408558	Key 1	Black	Round	3164708
Timing rail	3408509	3408381	3408585	Key 1	Black	Round	3164708

Other product changes with the QSK60 Phase 2.0 Harness Introduction:

- Exhaust gas temperature sensor improvements
- Improved breather design includes relocation
- Revise oil fill and dipstick arrangement
- Left bank oil level sensor relocation
- Pan adapter boss for oil level sensor
- Integral valve fan clutch control
- Relocation of Centinel™ valve to rear of engine on left bank side.

Note : The Phase 2.0 Harness can only be used in conjunction with the new crankcase breather design.

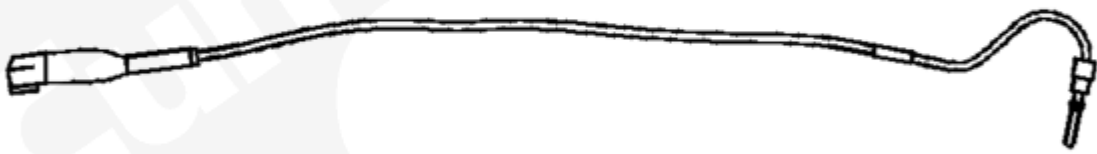
Harness Phase	Exhaust Gas Temperature Sensor Connector	Lead Length and Routing
Phase 1.5	Packard	1.2 m [4 ft] around intake manifold
Phase 2.0	Deutsch	Side port location with shortened lengths. Three exhaust gas temperature sensors bundled per connector

Exhaust Gas Temperature Sensor Improvements

- Redesigned to route between cylinder heads

- Changed cylinder head port location for improved serviceability
- Reduced length and strengthened wall section of Inconel portion
- Improved Deutsch type connector with stabilization
- Field upfit kit available.

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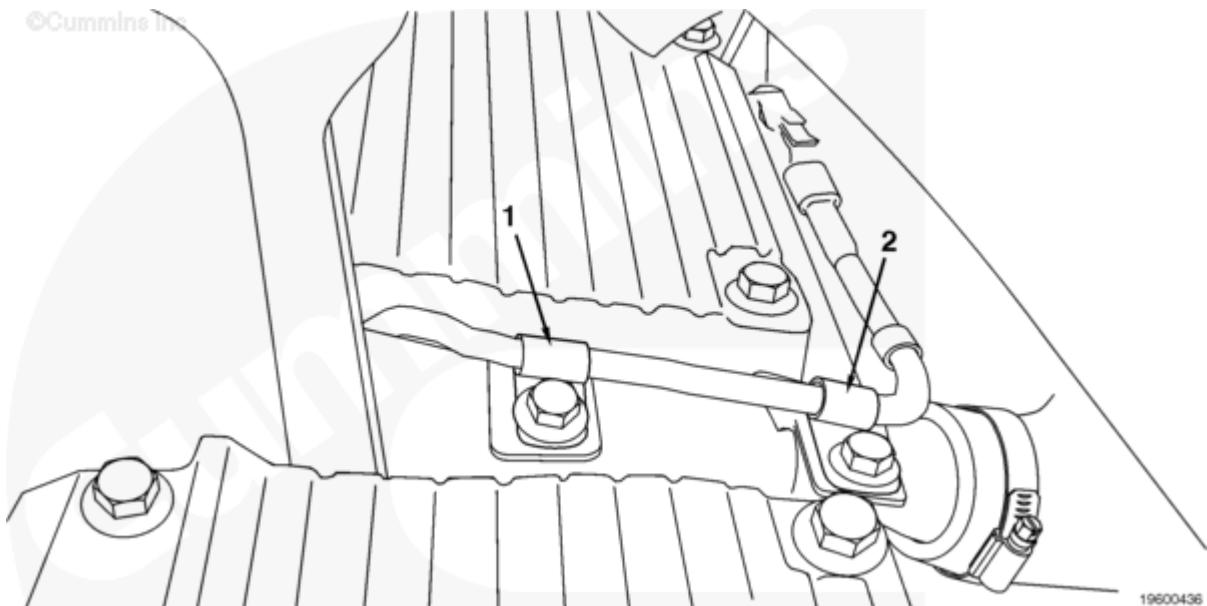
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Improved design is braided steel covered by a silicone sheath with Deutsch™ connector and heat shrink seal.



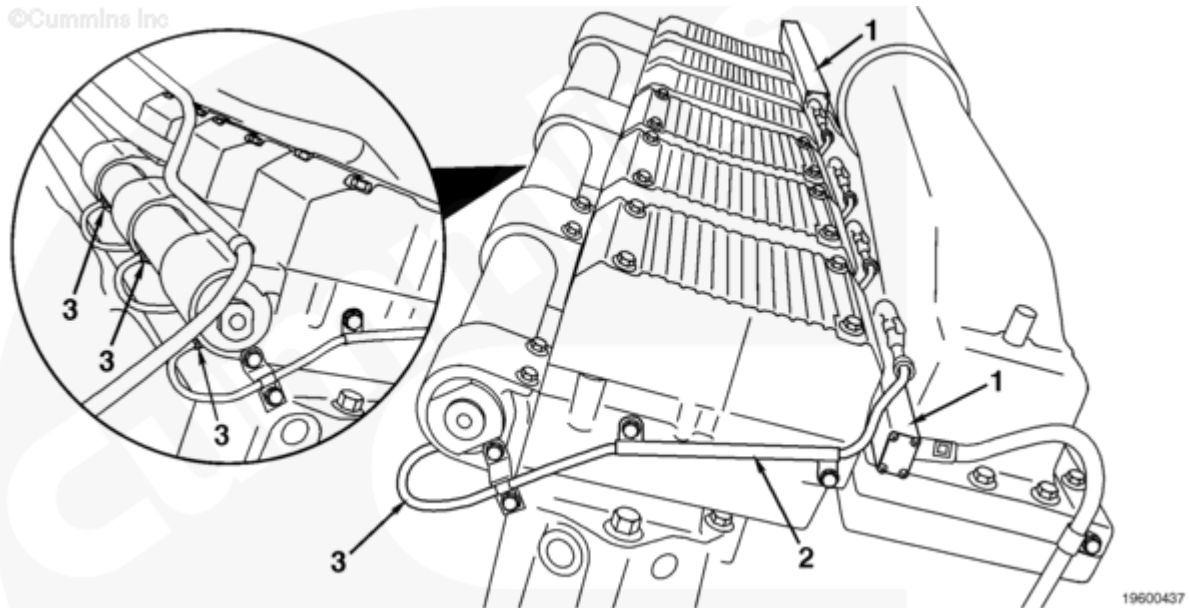
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The old Exhaust Gas Temperature Sensor, Inconel™ design.



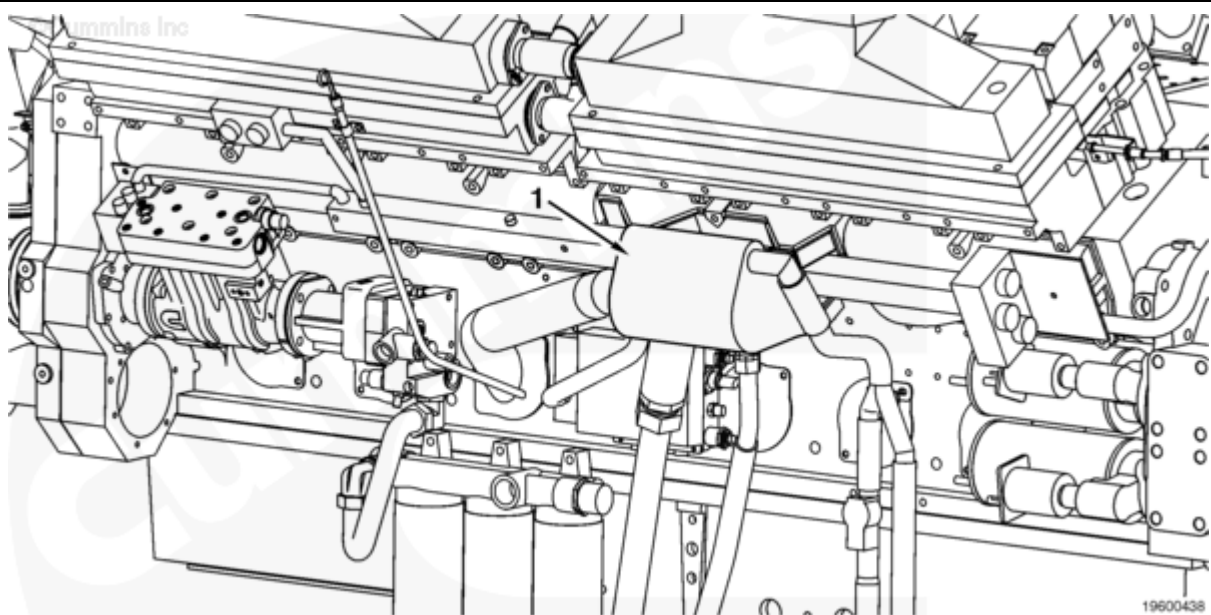
Phase 2.0 Exhaust Gas Temperature Sensor Routing

1 and 2 Mounting clips.



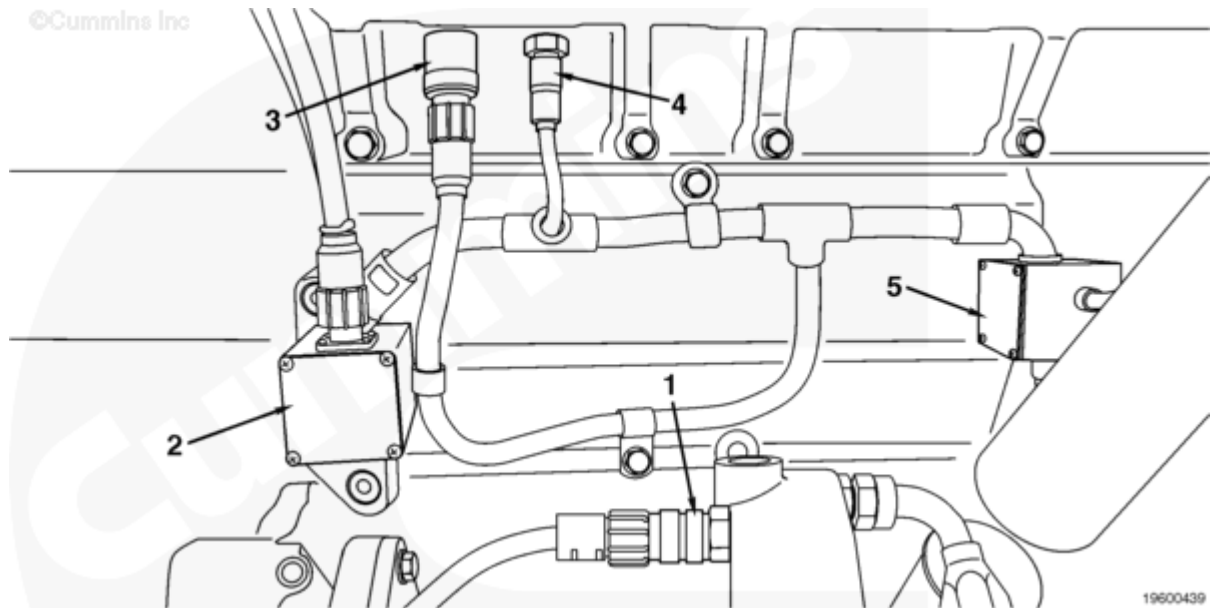
Exhaust Gas Temperature Sensor

1. Exhaust gas temperature sensor harness
2. Exhaust gas temperature sensor tunnel bracket
3. Exhaust gas temperature sensors.



Left Bank Engine

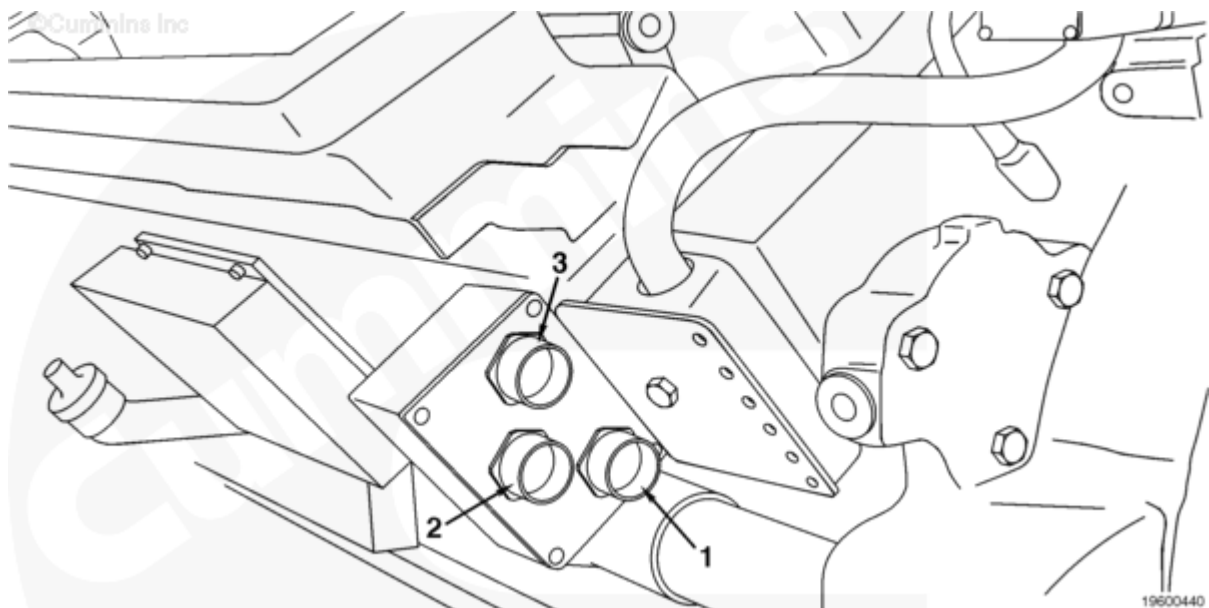
1. Crankcase breather.



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Left Bank Front Engine

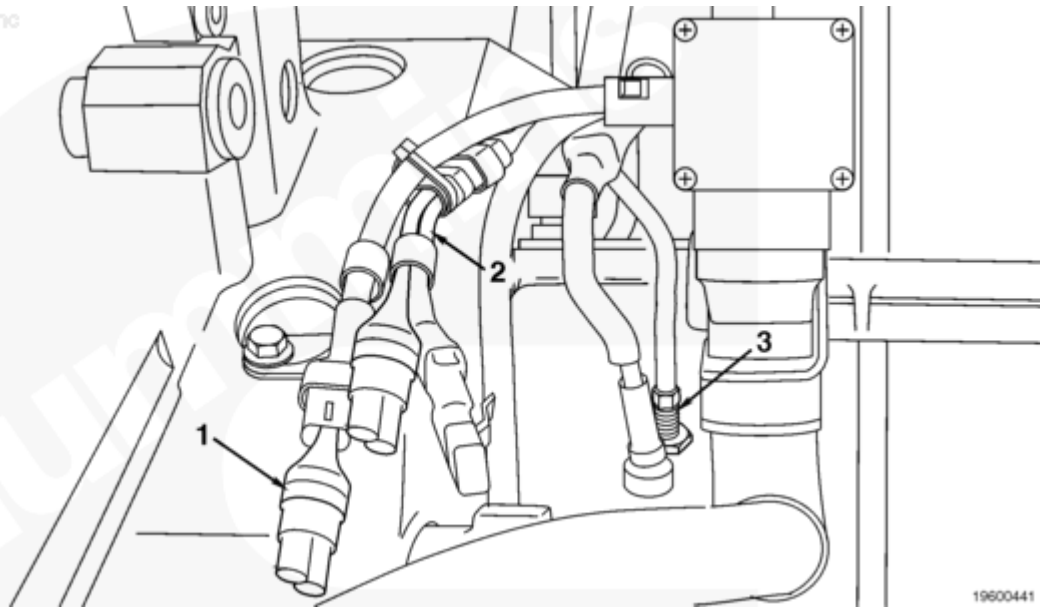
1. Fuel pump pressure sensor
2. Left bank lower front harness
3. Intake manifold pressure sensor
4. Intake manifold temperature sensor
5. Left bank middle harness.



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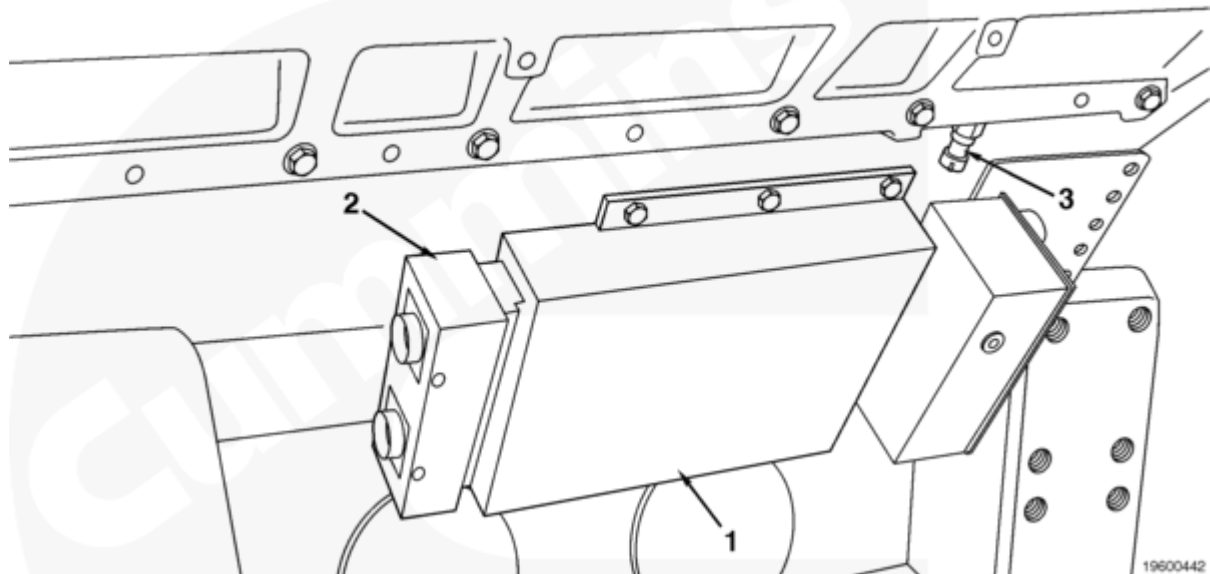
Left Bank Rear Corner Original Equipment Manufacturer Connectors.

1. 31-pin original equipment manufacturer cab disconnect
2. 21-pin original equipment manufacturer power sensor disconnect
3. 23-pin CENSE™ original equipment manufacturer disconnect.



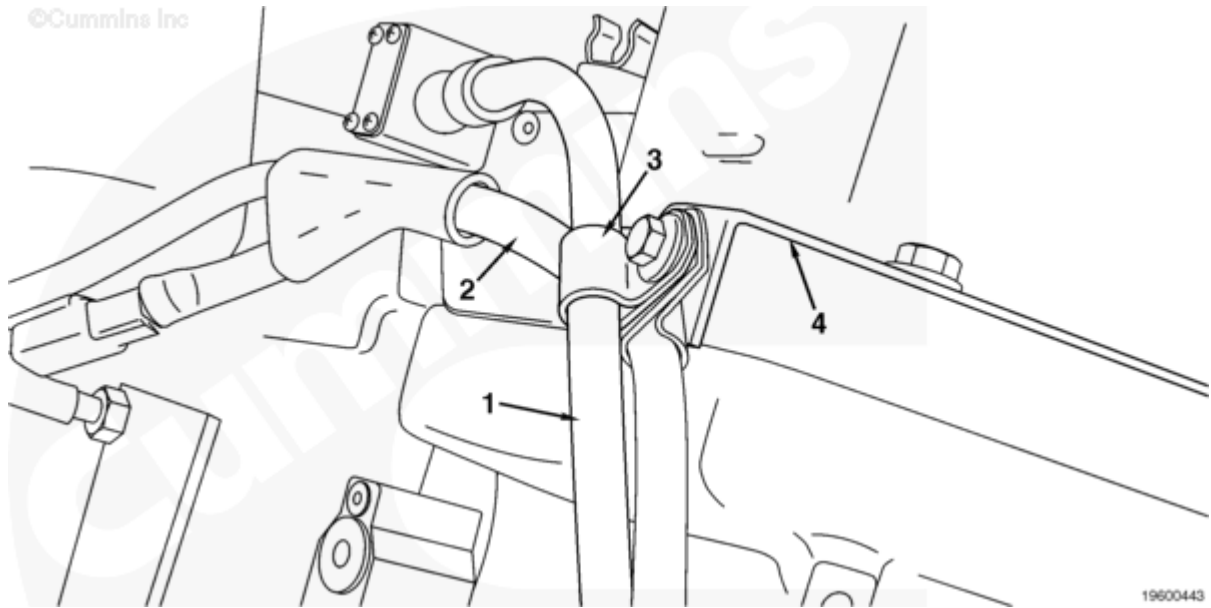
Left Bank Engine

1. Centinel™ Connection
2. Adaptor harness (single plug to two plugs) for original equipment manufacturer speed sensor.
3. Original equipment manufacturer speed sensor.



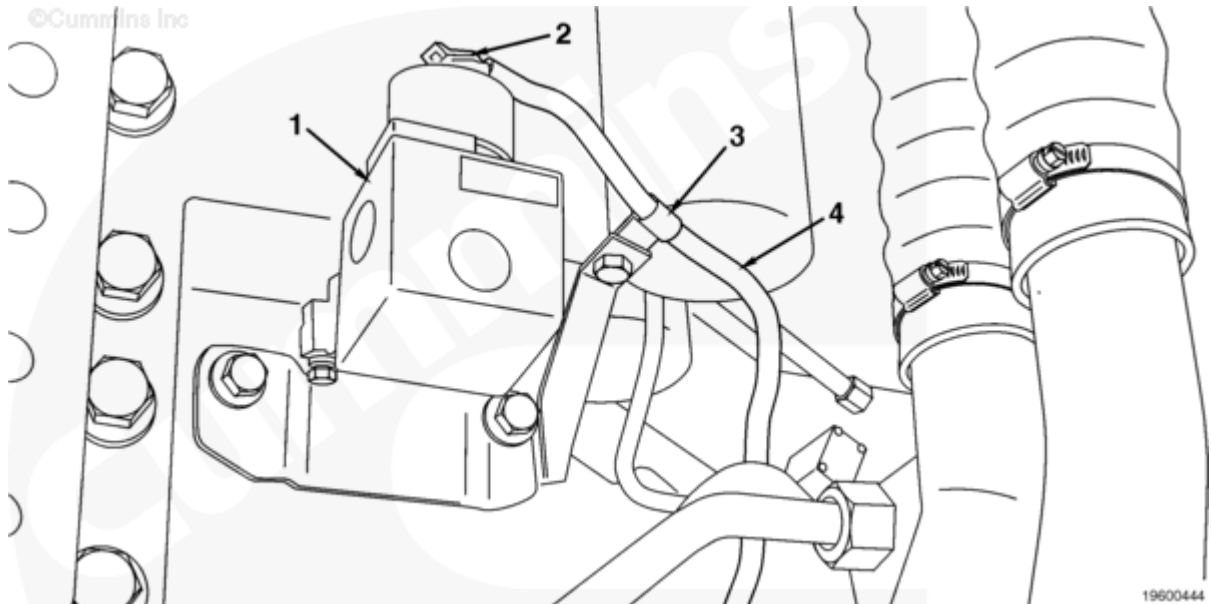
Left Bank Side of Engine (CENSE™ Module)

1. CENSE™ module cover bracket
2. CENSE™ Connector
3. Left bank rear intake manifold temperature sensor (CENSE™ sensor).



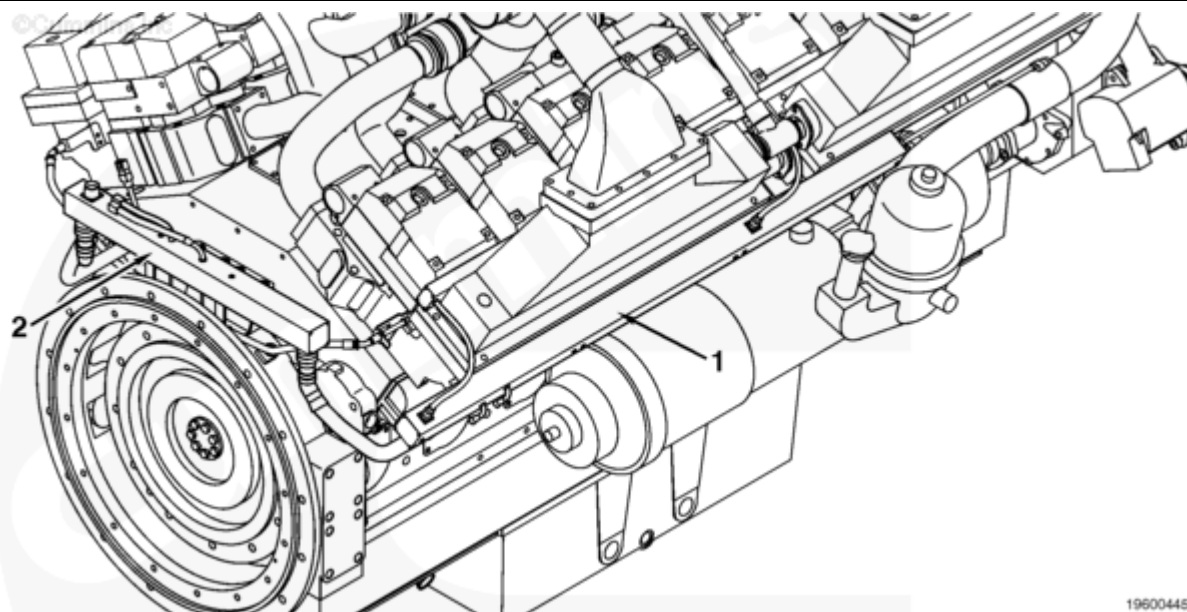
Left Bank Front

1. Exhaust Gas Temperature Sensor breakout (CENSE™ option only)
2. Fan/clutch coolant temperature sensor breakout
3. P-clips
4. Bracket.



Fan Clutch

1. Fan clutch actuator
2. Fan clutch connector
3. P-clips
4. Fan clutch extension harness.



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Rear of QSK60 2-Stage Engine

1. Right bank harness
2. Rear harness

Service Kits for QSK60 Engines with Head Deck Breather System and Phase 1.0 Harness

	Desired Harness						
Desired Breather		60-A Phase 1.0 without CENSE™	60-B Phase 1.0 with CENSE™	60-C Phase 1.5 without CENSE™	60-D Phase 1.5 with CENSE™	60-E Phase 2.0 without CENSE™	60-F Phase 2.0 with CENSE™
60-G Head deck	Not applicabl e	Not applicabl e	Kit 60-C	Kit 60-D	Not applicabl e	Not applicabl e	
60-H Single bank	Kit 60-H	Kit 60-H	Kits 60-C and 60-H	Kits 60-D and 60-H	Not applicabl e	Not applicabl e	
60-J Producti on/phas e 2 torpedo	Not applicabl e	Not applicabl e	Not applicabl e	Not applicabl e	Kits 60-E and 60-J	Kits 60-F and 60-J	

Service Kits for QSK60 Engines with Single Bank Breather System and Phase 1.0 Harness							
	Desired Harness						
Desired Breather		60-A Phase 1.0 without CENSE™	60-B Phase 1.0 with CENSE™	60-C Phase 1.5 without CENSE™	60-D Phase 1.5 with CENSE™	60-E Phase 2.0 without CENSE™	60-F Phase 2.0 with CENSE™
60-K Head deck	Not applicabl e	Not applicabl e	Not applicabl e	Not applicabl e	Not applicabl e	Not applicabl e	
60-L Single bank	Not applicabl e	Not applicabl e	Kit 60-C	Kit 60-D	Not applicabl e	Not applicabl e	
60-N Producti on/phas e 2 torpedo	Not applicabl e	Not applicabl e	Not applicabl e	Not applicabl e	Kits 60-E and 60-N	Kits 60-F and 60-N	

Service Kits for QSK60 Engines with Single Bank Breather System and Phase 1.5 Harness							
	Desired Harness						
Desired Breather		60-A Phase 1.0 without CENSE™	60-B Phase 1.0 with CENSE™	60-C Phase 1.5 without CENSE™	60-D Phase 1.5 with CENSE™	60-E Phase 2.0 without CENSE™	60-F Phase 2.0 with CENSE™
60-K Head deck	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	
60-L Single bank	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	
60-N Production/phase 2 torpedo	Not applicable	Not applicable	Not applicable	Not applicable	Kits 60-E and 60-N	Kits 60-F and 60-N	

Service Kits for QSK60 Engines with Deck Head Breather System and Phase 1.5 Harness							
	Desired Harness						
Desired Breather		60-A Phase 1.0 without CENSE™	60-B Phase 1.0 with CENSE™	60-C Phase 1.5 without CENSE™	60-D Phase 1.5 with CENSE™	60-E Phase 2.0 without CENSE™	60-F Phase 2.0 with CENSE™
60-K Head deck	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	
60-L Single bank	Not applicable	Not applicable	Kit 60-L	Kit 60-L	Not applicable	Not applicable	
60-J Production/phase 2 torpedo	Not applicable	Not applicable	Not applicable	Not applicable	Kits 60-E and 60-J	Kits 60-F and 60-J	

This kit is designed for the QSK60 engines phase 1.5 harness without CENSE™. It contains harnesses, sensors, and mounting hardware.

Kit 60-C Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 Harness Without CENSE™

Part Number	Name	Quantity	Remarks
69911	Clip	3	Two clips to support the oil level sensor and blowby sensor harness breakouts to the hand hole cover (left bank 3)
Supports a brace mounted on oil pan rail			
One clip to secure fuel pump pressure sensor and actuator harness breakouts to the left bank 1 and 2 cam cover at the lower capscrew location			
3093956	Screw, hexagonal flange head capscrew	1	M8 mounts p-clip holding oil rifle pressure sensor breakout to the support bracket at the rear of the engine.
3171395	Clip	4	Plastic p-clips securing electronic control module harness breakouts to vertically mounted electronic control module bracket.
3178520	Clip	2	Secures boost pressure, intake air temperature, or fan clutch breakouts to left bank 1 and 2 cam cover at the upper left and lower middle capscrew locations.

Kit 60-C Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 Harness Without CENSE™

Part Number	Name	Quantity	Remarks
3332136	Union, male	1	Male union fitting for coolant hose.
3408507	Adapter, electric connector	1	Danfoss Sensor - Packard connector adapter harness for boost pressure sensor.
3408509	Adapter, electric connector	1	Danfoss Sensor - Packard connector adapter harness for coolant pressure sensor.
3408509	Adapter, electric connector	1	Danfoss Sensor - Packard connector adapter harness for timing rail pressure sensor.
3408509	Adapter, electric connector	1	Danfoss Sensor - Packard connector adapter harness for fuel pump pressure sensor.
3408510	Adapter, electric connector	1	Danfoss Sensor - Packard connector adapter harness for fuel rail pressure.
3408511	Adapter, electric connector	1	Danfoss Sensor - Packard connector adapter harness for oil rifle pressure sensor.
3408556	Sensor, pressure	2	Danfoss boost pressure sensor (left bank front).
Coolant pressure sensor.			
3408562	Sensor, pressure	1	Danfoss fuel rail pressure sensor electronic control valve assembly.

Kit 60-C Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 Harness Without CENSE™

Part Number	Name	Quantity	Remarks
3408564	Sensor, pressure	1	Danfoss timing rail pressure sensor electronic control valve assembly.
3408566	Sensor, pressure	1	Danfoss fuel pump pressure sensor - remove snubber hose and snubber hose mountings from fuel pump.
3410392	Elbow, male union	1	Number 20 fuel filter head outlet fitting.
3628553	Brace, tube	1	Brace to support the oil level sensor and blowby sensor harness breakouts mounted on the oil pan rail.
3629452	Clip	4	Secures oil level sensor extension harness to oil pan.
Secures fuel pump actuator harness breakout to bottom of fuel pump flange.			
Secures fuel pump pressure sensor breakout to top of fuel pump flange.			
3630805	Clamp, wire tie	4	Secures fuel pump sensor and actuator breakouts together.
Secures the fuel pump sensor and actuator breakouts and the coolant pressure sensor breakout together.			

Kit 60-C Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 Harness Without CENSE™

Part Number	Name	Quantity	Remarks
Secures the fuel temperature and fuel pressure sensor electronic control valve assembly breakouts to the upper horizontal electronic control module bracket.			
Secures engine speed sensor breakouts and oil rifle pressure sensor breakout together.			
3657731	Clip	1	Secures oil rifle pressure sensor breakout to bracket at rear of engine.
3900589	Nut, hexagon flange	1	M8 secures p-clip to oil level sensor/blowby sensor support brace.
3900630	Screw, hexagonal flange capscrew	1	M8 secures p-clip to oil level sensor/blowby sensor support brace.
3900634	Screw, hexagonal flange capscrew	2	M10 secures enclosure bracket to cam covers.
3900679	Screw, hexagonal flange capscrew	1	M10 secures enclosure bracket to cam covers.
3901249	Screw, hexagonal flange capscrew	1	M12 secures oil level sensor/blowby sensor support brace to the oil pan rail.

Kit 60-C Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 Harness Without CENSE™

Part Number	Name	Quantity	Remarks
3906216	Nut, hexagon flange	5	M6 secures plastic p-clips to vertically mounted electronic control module bracket.
Secures weatherpack clips, holding speed sensor breakouts, to bracket mounted at the rear of the engine.			
Secures p-clip to fuel rail bracket at the front left of the engine.			
3913372	Screw, hexagonal flange capscrew	2	M12 secures speed sensor/oil pressure sensor bracket of the engine block.
3913638	Screw, hexagonal flange capscrew	3	M6 secure vertically mounted electronic control module harness support bracket to the horizontal.
Used to secure weatherpack clips to the support bracket at the rear of the engine.			
4000659	Harness, wiring	1	Extension harness to connect to oil level sensor located on the right bank side.
4007074	Elbow, male union	1	Fuel pump number 20 inlet fitting.

Kit 60-C Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 Harness Without CENSE™

Part Number	Name	Quantity	Remarks
4016678	Brace, wiring	1	Vertically mounted electronic control module harness breakout support bracket.
4017143	Bracket, enclosure	1	Brace to support interconnect box at the rear end of the left bank harness to cam cover.
4017568	Brace, wiring	2	Horizontal Electronic control module brackets.
4066054	Bracket, wiring retainer	1	Brace to support p-clips for harness breakouts for speed and oil rifle pressure sensor.
4066064	Clip	2	Weatherpack clips to support harness breakouts for speed sensor.
4066168	Clip	1	Secures fan clutch/coolant temperature sensor harness breakouts to the stud on the fuel rail bracket at left bank front of the engine.
4086499	Harness, external control module	1	Left bank vehicle and engine harness.
4088734	Sensor, pressure	1	Oil rifle pressure sensor (transparent gauge 0 to 689 kPa [0 to 100 psig] mounted on flywheel housing.
4100627	Hose, flexible	1	Coolant pressure sensor hose adapter.

Kit 60-C Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 Harness Without CENSE™

Part Number	Name	Quantity	Remarks
4100631	Adapter, o-ring	1	Adapter fitted into cylinder block in the kim-hot start location.
4101147	Hose, flexible	1	Number 20 fuel filter hose - fuel pump hose with 90 degree elbow.
QSK60 Engines Single Cylinder Air Compressors			
4100720	Hose, flexible	1	Fuel pump - electronic control valve assembly fuel hose.
QSK60 Engines with Twin Cylinder Air Compressors			
4080587	Hose, flexible	1	Fuel pump - electronic control valve assembly fuel hose.
QSK60 Engines without An Air Compressor			
4065606	Hose, flexible	1	Fuel pump - electronic control valve assembly fuel hose.
QSK60 Engines With "00" Flywheel Housing			
3408471	Sensor, speed	1	Four-pin single plug speed sensor for "00" flywheel housing.
3408526	Adapter, electrical connector	1	Deutsch connector - Packard connector adapter harness for engine speed sensor.

Kit 60-C Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 Harness Without CENSE™			
Part Number	Name	Quantity	Remarks
QSK60 Engines With "0" Flywheel Housing			
3408472	Sensor, speed	1	Four-pin single plug speed sensor for "0" flywheel housing.
3408526	Adapter, electrical connector		Deutsch connector - Packard connector adapter harness for engine speed sensor.
QSK60 Engines With Rockford Fan clutch Solenoid			
114307	Brace, tube	1	Brace to support the fan clutch breakout.
3093956	Screw, hexagon flange head capscrew	1	M8 secures p-clip holding fan clutch breakout to support brace.
3629452	Clip	1	Secures fan clutch breakout to fan clutch breakout support brace.
3900589	Nut, hexagon flange	1	M8 secures p-clip holding fan clutch breakout to support brace.
3903990	Screw, hexagon flange head capscrew	1	M10 mounts support brace to idler pulley bracket.

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
69911	Clip	2	Clip to support the oil level sensor and blowby sensor harness breakouts to the support brace mounted on the oil pan rail.
Clip to secure fuel pump sensor and actuator harness breakouts to the left bank plus 2 cam cover at the lower capscrew location.			
70709	Washer, plain	2	Used in conjunction with p-clip and nut on left bank curved CENSE™ harness brace.
Used in conjunction with p-clip and nut on right bank curved CENSE™ harness brace.			
180371	Clip	32	Used on each cylinder support exhaust gas temperature sensor extension harness.
Support exhaust gas temperature sensor extension harness on Z-bracket mounted under coolant transfer tube.			

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
3081825	Clip	4	One per quadrant used to secure all four exhaust gas temperature sensor extension harnesses in quadrant to aftercooler cover.
3093946	Nut, regular hexagon	8	M6 secures clip 4017498 to small studs on wiring harness brace and left bank front brace.
3093956	Screw, hexagonal flange capscrew	1	M8 mounts p-clip holding oil rifle pressure sensor to the support bracket at the rear of the engine.
3171395	Clip	4	Plastic p-clips securing electronic control valve assembly harness breakouts to vertically mounted electronic control module bracket.
3171451	Screw, pan head capscrew	36	Connects 4 spacers to the left bank front brace.
Connects 4 spacers to the left bank rear brace.			
Secures RS232 data link to flange on left bank front harness.			
Connects 4 spacers to the right bank front brace.			

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™

Part Number	Name	Quantity	Remarks
Connects 4 spacers to the right bank rear brace.			
3171470	Washer, plain	4	Used on RS232 data link support flange on the left bank front brace.
3178520	Clip	3	Secures boost pressure/intake air temperature/fan clutch breakouts to left bank 1 cam cover upper left and lower middle capscrew locations.
Secures oil temperature/oil level/blowby flow harness breakouts to left bank 6 hand hole cover.			
3332136	Union, male	1	Male union fitting for coolant hose.
3408507	Adapter, electric connector	2	Danfoss Sensor - Packard connector adapter harness boost pressure sensor (left bank front and right bank rear).
3408509	Adapter, electric connector	5	Danfoss Sensor - Packard connector adapter harness coolant pressure sensor.
Lubrication pressure sensors.			

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
Timing rail pressure sensor electronic control valve assembly.			
Fuel pump pressure sensor.			
3408510	Adapter, electric connector	12	Danfoss Sensor - Packard connector adapter harness for fuel rail pressure sensor electronic control valve assembly.
3408511	Adapter, electric connector	1	Danfoss Sensor - Packard connector adapter harness for oil rifle pressure sensor.
3408540	Sensor, exhaust gas temperature	16	Temperature sensors.
3408556	Sensor, pressure	3	Danfoss Sensor - boost pressure sensor (left bank front and right bank rear).
Coolant pressure sensor.			
3408558	Sensor, pressure	2	Danfoss pre and post lube oil pressure sensors.
3408562	Sensor, pressure	1	Danfoss fuel rail pressure sensor electronic control valve assembly.
3408564	Sensor, pressure	1	Danfoss timing rail pressure sensor electronic control valve assembly.
3408566	Sensor, pressure	1	Danfoss fuel pump pressure sensor.

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
3410392	Elbow, male union	1	Number 20 fuel filter head outlet fitting.
3628553	Brace, tube	1	Brace to support the oil level/temperature sensors and the blowby sensor harness breakouts mounted on the oil pan rail.
369451	Clip	2	Mounts CENSE™ harness to left bank curved CENSE™ harness brace.
Mounts CENSE™ harness to right bank curved CENSE™ harness brace.			
3629452	Clip	5	Secures oil level sensor extension harness to oil pan.
Secures fuel pump actuator harness breakout to bottom of fuel pump flange.			
Secures fuel pump pressure sensor breakout to top of fuel pump flange.			
Secures oil temperature CENSE™ harness breakout to the support on the oil pan rail.			
3630804	Clamp, wire tie	3	Secures backbone harness to the CENSE™ harness on main brace.
3630805	Clamp, wire tie	40	Secures fuel pump sensor and actuator breakouts together.

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
Secures the fuel pump sensor and actuator breakouts and the coolant pressure sensor breakout together.			
Secures the fuel temperature and fuel pressure sensor breakouts to the upper horizontal electronic control module brackets.			
Secures engine speed sensor breakouts and oil rifle pressure sensor together.			
Secures backbone harness to CENSE™ harness on left bank curved CENSE™ harness brace.			
Secures exhaust gas temperature sensor extension harness to the CENSE™ harness (2 for each cylinder) .			
3657731	Clip	1	Secures oil rifle pressure sensor breakout to bracket at rear of engine.
3823336	Kit, butt splice	32	Connects the exhaust gas temperature sensor extension harness to the CENSE™ harness.

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
3900589	Nut, hexagon flange	1	M8 secure p-clip to oil level sensor/blowby sensor support brace mounted on the oil pan rail.
3900629	Screw, hexagonal flange capscrew	4	M8 secures the right bank curved bracket to the harness brace.
Secures the left bank curved bracket to the harness brace.			
3900629	Screw, hexagonal flange capscrew	20	M8 secures the left bank front CENSE™ harness brace to the intake manifold (upper).
Secures the left bank rear CENSE™ harness brace to the intake manifold (upper).			
Secures the right bank front CENSE™ harness brace to the intake manifold (upper).			
Secures the right bank rear CENSE™ harness brace to the intake manifold (upper).			
Secures the exhaust gas temperature sensor Z-bracket to the rocker housing.			
Secures the p-clips to the Z-bracket.			

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
3900630	Screw, hexagonal flange capscrew	1	M8 secure p-clip to oil level sensor/blowby sensor support brace mounted on the oil pan rail.
3900634	Screw, hexagonal flange capscrew	2	M10 secure enclosure bracket to cam cover.
3900679	Screw, hexagonal flange capscrew	1	M10 secures enclosure bracket to cam covers.
3901249	Screw, hexagonal flange capscrew	1	M12 secures oil level sensor/blowby sensor support brace to the oil pan rail.
3901445	Screw, hexagonal flange capscrew	16	M8 secures the left bank front CENSE™ harness brace to the intake manifold (lower).
Secure the left bank rear CENSE™ harness brace to the intake manifold (lower).			
Secures the right bank front CENSE™ harness brace to the intake manifold (lower).			
Secures the right bank front CENSE™ harness brace to the intake manifold (lower).			
3902460	Screw, hexagonal flange capscrew	8	M10 secure the right bank curved bracket to the support bracket.

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
Secures the left bank curved bracket to the support bracket.			
Secures the support brackets for the curved brace to the flanges on the CENSE™ module bracket and mount the heat shield.			
3903990	Screw, hexagonal flange capscrew	4	M10 secure the CENSE™ module housing to the mounting brackets.
3906216	Nut, hexagon flange	20	M6 securing plastic p-clips to vertically mounted electronic control module bracket.
Secures weatherpack clips, holding speed sensor breakouts, to bracket.			
Secures p-clip to fuel rail bracket mounted at the front left of the engine.			
Left bank front CENSE™ harness brace.			
Left bank rear CENSE™ harness brace.			
Right bank front CENSE™ harness brace.			
Right bank rear CENSE™ harness brace.			

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
Secures T-section flange to left bank curved CENSE™ harness brace.			
Secures p-clip to left bank curved CENSE™ harness brace.			
Secures T-section flange to right bank curved CENSE™ harness brace.			
Secures p-clip to right bank curved CENSE™ harness brace.			
3913372	Screw, hexagonal flange capscrew	2	M12 secures speed sensor/oil pressure sensor bracket to the engine block.
3913638	Screw, hexagonal flange capscrew	3	M6 secure vertically mounted electronic control module harness support bracket to the horizontal electronic control module brackets.
Used to secure weatherpack clips to the support bracket at the rear of the engine.			
4000659	Harness, wiring	1	Extension harness to connect to oil level sensor mounted on the right bank side.
4007074	Elbow, male union	1	Fuel pump number 20 inlet fitting.

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
4016678	Brace, Wiring	1	Vertically mounted electronic control valve assembly harness breakout support bracket.
4017143	Bracket, enclosure	1	Brace to support interconnect box at the rear end of the left bank harness.
4017464	Harness, external control middle wiring	1	Right bank CENSE™ harness wiring.
4017465	Harness, external control middle wiring	1	Left bank CENSE™ harness wiring.
4017483	Brace, wiring	1	right bank curved brace, supporting CENSE™ harness to the CENSE™ module.
4017498	Clip	8	Holds clip 4017499 left bank front brace.
Right bank front brace.			
4017499	Clip	8	Supports CENSE™ harness breakouts on harness braces (attached to clip 4017498).
Left bank front brace.			
Right bank front brace.			
4017514	Harness, control middle assembly	1	CENSE™ electronic control module harness, housing, cover, and assembly.
4017519	Harness, wiring	1	Backbone extension harness.
4017568	Brace, wiring	2	Horizontal electronic control module brackets.

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
4017634	Brace, wiring	2	Curved brace support brackets, mounted to flanges on CENSE™ electronic control module housing support brackets.
4065922	Spacer, mounting	16	Spacer placed between harness support braces and the intake manifold.
Left bank front harness support brace.			
Left bank rear harness support brace.			
Right bank front harness support brace.			
Right bank rear harness support brace.			
4066053	Shield, heat	1	Mounted on the back of the CENSE™ module, on the flanges on the CENSE™ electronic control module housing support brackets.
4066054	Bracket, wiring retainer	1	Brace to support p-clips for harness breakouts for speed and oil rifle pressure sensor.
4066064	Clip	2	Clips to support harness breakouts for speed sensor.

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™

Part Number	Name	Quantity	Remarks
4066070	Brace, wiring	1	Left bank rear CENSE™ harness support brace mounted on intake manifolds.
4066144	Brace, wiring	1	Right bank rear CENSE™ harness support brace mounted on intake manifolds.
4066148	Brace, wiring	1	Left bank front CENSE™ harness support brace mounted on intake manifolds.
4066168	Clip	1	Secures fan clutch/coolant temperature sensor to fuel rail bracket at left bank front of the engine.
4066560	Z-bracket	16	Mounts p-clip onto the rocker box capscrew under the low temperature aftercooler coolant transfer tube.
4066813	Harness, exhaust gas temperate extension	16	Extension harness for exhaust gas temperature sensor.
4080161	Brace, wiring	1	Right bank front CENSE™ harness support brace mounted on intake manifolds.
4086499	Harness, electronic control module	1	Left bank vehicle and engine harness.

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
4088734	Sensor, pressure	1	Oil rifle pressure sensor (transparent gauge 0 to 689 kPa [0 to 100 psi] mounted on flywheel housing.
4100464	Brace, wiring	1	Left bank curved brace, supporting CENSE™ harness to the CENSE™ module.
4100627	Hose, flexible	1	Coolant pressure sensor hose adapter.
4100631	Adapter, o-ring	1	Adapter fitted into cylinder block in the kim-hot start location.
4101147	Hose, flexible	1	Number 20 fuel filter hose - fuel pump hose with 90 degree elbow.
QSK60 Engines with Single Cylinder Air Compressors Add to Kit 60-D			
4100720		1	Fuel pump - electronic control valve assembly fuel hose.
QSK60 Engines with Twin Cylinder Air Compressors Add to Kit 60-D			
4080587		1	Fuel pump - electronic control valve assembly fuel hose.
QSK60 Engines Without an Air Compressors Add to Kit 60-D			

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
4065606	Hose, flexible	1	Fuel pump - electronic control valve assembly fuel hose.
QSK60 Engines With "00" Flywheel Housing Add to Kit 60-D			
3408471	Sensor, speed	1	Four pin single plug speed sensor for "00" flywheel housing.
3408526	Adapter, electric connector	1	Deutsch connector - Packard connector adapter harness for engine speed sensor.
3903990	Screw, hexagonal flange head capscrew	4	Mounts support brackets to flywheel housing and engine block.
4066097	Bracket, module	1	Rear left CENSE™ module support bracket.
4066104	Bracket, module	1	Rear right CENSE™ module support bracket.
QSK60 Engines With "0" Flywheel Housing Add to Kit 60-D			
3408472	Sensor, speed	1	Four pin single plug speed sensor for "0" flywheel housing.
3408526	Adapter, electric connector	1	Deutsch connector - Packard connector adapter harness for engine speed sensor.
3903990	Screw, hexagonal flange head capscrew	4	Mounts support brackets to flywheel housing and engine block.

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
4066081	Bracket, module	1	Rear left CENSE™ module support bracket.
4066089	Bracket, module	1	CENSE™ module support bracket.
QSK60 Engines With Rockford Fan Clutch Solenoid Add to Kit 60-D			
114307	Brace, tube	1	Brace to support the fan clutch breakout.
3093956	Screw, hexagonal flange head capscrew	1	M8 secures p-clip to fan clutch breakout support brace.
3629452	Clip	1	Secures fan clutch breakout to fan clutch breakout support brace.
3900589	Nut, hexagonal flange	1	M8 secures p-clip to fan clutch breakout support brace.
3903990	Screw, hexagonal flange head capscrew	1	M10 mounts support brace to the idler pulley bracket.
QSK60 Engines Without Eliminator Add to Kit 60-D			
3629452	Clip	4	Secures pre/post oil filter pressure sensor CENSE™ harness breakouts to the cam cover and oil filter head.
M8 secure the right bank front protective cover to the harness brace.			

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
3900629	Screw, hexagonal flange head capscrew	24	Secures the right bank rear protective cover to the harness brace.
Secures the left bank rear protective cover to the harness brace.			
Secures the left bank front protective cover to the harness brace.			
4017479	Cover, protective	1	Left bank rear CENSE™ harness protective cover.
4017493	Cover, protective	1	Right bank rear CENSE™ harness protective cover.
4017494	Cover, protective	1	Right bank front CENSE™ harness protective cover.
4066130	Cover, protective	1	Left bank front CENSE™ harness protective cover.
QSK60 Engines With Eliminator Add to Kit 60-D			
3630805	Clamp, wire tie	4	Secures eliminator post/pre pressure sensor CENSE™ breakouts.
3900629	Screw, hexagonal flange head capscrew	24	M8 secure the right bank front protective cover to the harness brace.
Secures the right bank rear protective cover to the harness brace.			

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
Secures the left bank rear protective cover to the harness brace.			
Secures the left bank front protective cover to the harness brace.			
4017479	Cover, protective	1	Left bank rear CENSE™ harness protective cover.
4017493	Cover, protective	1	Right bank rear CENSE™ harness protective cover.
4066130	Cover, protective	1	Right bank front CENSE™ harness protective cover.
4080162	Cover, protective	1	Left bank front CENSE™ harness protective cover.
WA1200 Engine Add to Kit 60-D			
3629452	Clip	4	Secures pre/post oil filter pressure sensor CENSE™ harness breakouts to the cam cover and dummy oil filter head.
3900629	Screw, hexagonal flange head capscrew	24	M8 secure the right bank front protective cover to the harness brace.
Secures the right bank rear protective cover to the harness brace.			
Secures the left bank rear protective cover to the harness brace.			

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
Secures the left bank front protective cover to the harness brace.			
4017479	Cover, protective	1	Left bank rear CENSE™ harness protective cover.
4066181	Cover, protective	1	Right bank rear CENSE™ harness protective cover.
4066182	Cover, protective	1	Right bank front CENSE™ harness protective cover.
4066183	Cover, protective	1	Left bank front CENSE™ harness protective cover.
Komatsu Engine Without Eliminator Add to Kit 60-D			
3629452	Clip	4	Secures pre/post oil filter pressure sensor CENSE™ harness breakouts to the cam cover and oil filter head.
3900629	Screw, hexagonal flange head capscrew	24	M8 secure the right bank front protective cover to the harness brace.
Secures the right bank rear protective cover to the harness brace.			
Secures the left bank rear protective cover to the harness brace.			
Secures the left bank front protective cover to the harness brace.			

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
4017479	Cover, protective	1	Left bank rear CENSE™ harness protective cover.
4017493	Cover, protective	1	Right bank rear CENSE™ harness protective cover.
4017494	Cover, protective	1	Right bank front CENSE™ harness protective cover.
4066183	Cover, protective	1	Left bank front CENSE™ harness protective cover.
Komatsu Engine With Eliminator Add to Kit 60-D			
3630805	Clamp, wire tie	4	Secures eliminator post/pre pressure sensor CENSE™ breakouts.
3900629	Screw, hexagonal flange head capscrew	24	M8 secure the right bank front protective cover to the harness brace.
Secures the right bank rear protective cover to the harness brace.			
Secures the left bank rear protective cover to the harness brace.			
Secures the left bank front protective cover to the harness brace.			
4017479	Cover, protective	1	Left bank rear CENSE™ harness protective cover.
4017493	Cover, protective	1	Right bank rear CENSE™ harness protective cover.

Kit 60-D Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 1.5 With CENSE™			
Part Number	Name	Quantity	Remarks
4066183	Cover, protective	1	Right bank front CENSE™ harness protective cover.
4080162	Cover, protective	1	Left bank front CENSE™ harness protective cover.
Kit 60-E Harnesses, Sensors and Mounting Hardware for QSK60 Engines Phase 2.0 Without CENSE™			
Part Number	Name	Quantity	Remarks
68152	Clip	1	Secures breakout for coolant temperature sensor and fan clutch solenoid to bracket on mounted on the left bank front after cooler cover.
68152 B	Clip	1	Secures blowby harness breakout to oil pan rail.
69465	Clip	1	Secures harness breakout that links middle and front section of the left bank harness to the cam follower cover at the upper capscrew location.
202888	Clip	1	P-clip to secure coolant pressure sensor hose to L-bracket on the left bank 1 hand hole cover.

3089316	Screw, hexagonal flange head capscrew	3	Secures quantum harness box to the cam follower covers at the left middle capscrew locations.
3093828	Nut, hexagonal head flange	1	M10 secures p-clip for coolant pressure sensor hose to L-bracket on the left bank 1 hand hole cover.
3093957	Screw, hexagonal flange capscrew	2	M10 mounts L-bracket for Centinal™ harness breakout to block at rear left of the engine.
Secures p-clip for coolant pressure sensor hose adapter to L-bracket mounted on the left bank 1 hand hole cover.			
3332136	Union, male	1	Male union for fitting coolant pressure sensor hose.
3408556	Sensor, pressure	2	Left bank front intake manifold (boost) pressure sensor.
3408562	Sensor, pressure	1	Fuel rail pressure sensor (part of electronic control valve assembly assembly).
3408564	Sensor, pressure	1	Timing rail pressure sensor (part of electronic control valve assembly assembly).
3408566	Sensor, pressure	1	Fuel pump pressure sensor.
3410392	Elbow, male union	1	Number 20 fuel filter head outlet fitting.

3627412	Clip	1	Secures harness breakout for fuel pump pressure sensor and actuator to the left bank 1 and 2 cam follower cover at the lower capscrew location.
3628553	Brace, tube	2	L-bracket mounted to hand hole cover left bank 3 for mounting the Quantum electronic control module ground breakout.
L-bracket for securing coolant pressure sensor hose to the left bank 1 hand hole cover.			
368585	Brace tube	1	L-bracket to secure Centinal™/OEM speed sensor breakout to block at the rear left of the engine.
3900589	Nut, hexagonal head flange	2	M8 secures ground cable to L bracket on the left bank 3 hand hole cover.
Secures p-clip holding Centinal™ harness breakout to L-bracket on block rear left of the engine.			
3900629	Screw, hexagonal flange capscrew	2	M8 secures ground cable to L-bracket on block at rear left of the engine.
3900630	Screw, hexagonal flange capscrew	1	M8 mounts p-clip onto bracket on front left aftercooler cover.

3900679	Screw, hexagonal flange capscrew	3	M10 mounts harness box at the rear of the engine to the flywheel housing or support bracket.
Mounts to the front section of the quantum harness to front cam follower cover at lower capscrew location.			
3906216	Nut, hexagonal flange capscrew	1	M6 secures p-clip onto stud on the bracket on left bank front fuel rail.
3913372	Screw, hexagonal flange capscrew	1	M12 secures p-clip holding blowby breakout to the oil pan adapter.
4007074	Elbow, male union	1	Fuel pump number 20 inlet fitting.
4066935	Brace, wiring	1	Brace to secure fan clutch/coolant temperature harness breakout to the left bank front aftercooler cover.
4067054	Harness, electronic control module wiring	1	Engine harness, mounted on the left bank with No Cense™ option. Harness will have Packard spade fuses, sealed OEM box, potted back shells, Y's and T's and Schlemer conduit.
4086487	Harness, electronic control module wiring	1	Rear section of the engine harness for No Cense™ option (mounted on flywheel housing or support bracket on flywheel housing).

4088734	Sensor, pressure	1	Oil rifle pressure sensor (transparent gauge 0 to 689 kPa [0 to 100 psig]) mounted on flywheel housing.
4100552	Clip	12	Secures oil level sensor breakout on right bank to cam follower cover.
Lubricating filter head.			
Oil pan adapter.			
Secures Centinal™ harness breakout to L-bracket on block at the rear left of the engine.			
Secures fuel pump pressure sensor breakout to fuel pump flange.			
Secures coolant pressure sensor harness breakout to cam covers at the lower capscrew location.			
4100627	Hose, flexible	1	Coolant pressure sensor hose adapter.
4100631	Adapter, o-ring	1	Adapter fitted into cylinder block in the kim-hot start location.
4101147	Hose, flexible	1	Number 20 fuel filter hose - fuel pump hose with 90 degree elbow.
QSK60 Engines With Single Cylinder Air Compressors Add to Kit 60-E			

4100720	Hose, flexible	1	Fuel pump electronic control valve assembly fuel hose.
QSK60 Engines Twin Single Cylinder Air Compressors Add to Kit 60-E			
4080587	Hose, flexible	1	Fuel pump electronic control valve assembly fuel hose.
QSK60 Engines Without An Air Compressor Add to Kit 60-E			
4065606	Hose, flexible		Fuel pump electronic control valve assembly fuel hose.
QSK60 Engines With "00" Flywheel Housing Add to Kit 60-E			
3408471	Sensor, speed	1	Four pin single plug speed sensor for "00" flywheel housing.
QSK60 Engines With "0" Flywheel Housing Add to Kit 60-E			
3408472	Sensor, speed	1	Four pin single plug speed sensor for "0" flywheel housing.
3902460	Screw, hexagonal flange capscrew	2	M10 mounts bracket onto the flywheel housing.
4067011	Bracket, external control module wiring	1	Bracket to mount rear section of the harness box onto the flywheel housing.
QSK60 Engines With Rockford Fan Clutch Solenoid Add to Kit 60-E			

114307	Brace, tube	1	Brace to support the fan clutch breakout extension harness.
3093956	Screw, hexagonal flange head capscrew	1	M8 secures p-clip to fan clutch breakout support brace.
3900589	Nut, hexagonal flange	1	M8 secures p-clip to fan clutch breakout support brace.
3629452	Clip	1	Secures fan clutch breakout to support brace.
3903990	Screw, hexagonal flange head capscrew	1	M10 mounts support brace to the idler pulley bracket.
4066664	Harness, electronic control module wiring	1	Fan clutch extension harness.
QSK60 Engines With Burn and Make-Up Centinal™ Oil Replenishing System Add to Kit 60-E			
3170823	Elbow, male union	1	
3682300	Union, male	1	
3900635	Screw, hexagonal flange head capscrew	2	
3901249	Screw, hexagonal flange head capscrew	2	
3903990	Screw, hexagonal flange head capscrew	3	
4000694	Connector, male	1	
4066833	Bracket, valve	1	
4086288	Adapter, o-ring	1	
4086294	Union, male	1	
4086355	Hose, flexible	1	
4086356	Hose, flexible	1	
4086357	Hose, flexible	1	

4086382	Clamp, retaining	1	
4086648	Valve, oil control	1	
QSK60 Engines With Burn Only Centinal™ Oil Replenishing System Add to Kit 60- E			
3170823	Elbow, male union	1	
3682300	Union, male	1	
3900635	Screw, hexagonal flange head capscrew	2	
3901249	Screw, hexagonal flange head capscrew	2	
3903990	Screw, hexagonal flange head capscrew	3	
4000694	Connector, male	1	
4016411	Elbow, male union	1	
4016412	Adapter, reducing	1	
4016565	Hose, flexible	1	
4066833	Bracket, valve	1	
4086294	Union, male	1	
4086355	Hose, flexible	1	
4086357	Hose, flexible	1	
4086648	Valve, oil control	1	

Kit 60-F Harnesses, Sensor and Mounting Hardware for QSK60 Engines Phase 2.0 With Cense™			
Part Number	Name	Quantity	Remarks
68152	Clip	1	Secures breakout for coolant temperature sensor and fan clutch solenoid to bracket on left bank front after cooler cover.
68152 B	Clip	1	Secures blowby harness breakout to oil pan rail.
69465	Clip	1	Secures harness breakout that links middle and front section of the left bank harness to the cam follower cover at the upper capscrew location.
108722	Clip	4	Secures exhaust gas temperature sensor harness to support brace on the right bank rear after cooler cover at the rear of the engine.
Secures exhaust gas temperature sensor harness to support brace on the left bank front after cooler cover at the front of the engine.			

Secures exhaust gas temperature sensor harness to support brace on the right bank front after cooler cover towards the middle of the engine.			
Secures exhaust gas temperature sensor harness to support brace on the left bank rear after cooler cover towards the middle of the engine.			
180371	Clip	20	Secures exhaust gas temperature sensor harness to Z-bracket on rocker housing (left bank 1 only).
Secures exhaust gas temperature sensor harnesses to rocker housing below water transfer tubes (all except left bank 1).			
Secures exhaust gas temperature sensor harness above water transfer tubes (left bank 5 and right bank 4).			
202888	Clip	1	P-clip to secures coolant pressure sensor hose to L-bracket on the right bank 1 hand hole cover.

3062329	Clamp, wire tie	1	secures exhaust gas temperature sensor breakout to coolant/fan clutch breakout at the left bank front or the engine.
3089316	Screw, hexagonal flange head capscrew	9	M10 secures quantum harness to the left bank cam follower covers at the left and middle capscrew locations.
Secures Cense™ harness box to the cam covers on the right bank of the engine.			
3093828	Nut, hexagonal head flange	1	M10 secures p-clip for coolant pressure sensor hose to L-bracket on the left bank 1 hand hole cover.
3093956	Screw, hexagonal flange head capscrew	49	M8 mounts the Z-bracket to the rocker housing below the water transfer tubes.
Secures p-clip to Z-bracket on rocker housing.			
Secures wiring brace to rocker housing (left bank 1).			
Secures p-clip to rocker housing (left bank 5 and right bank 4).			
Secures p-clips above water transfer tubes (left bank 5 and right bank 4).			

Secures p-clips to rocker housing below water transfer tubes.			
Secures wiring brace to rocker housing.			
3093957	Screw, hexagonal flange capscrew	2	M10 mounts L-bracket for Centinal™ harness breakout to block at rear left of the engine.
Secures p-clip for coolant pressure sensor hose adapter to L-bracket mounted on the left bank 1 hand hole cover.			
3331658	Connector, female	1	Adapter for left bank 1 exhaust gas temperature sensor harness.
3332136	Union, male	1	Male union fitting coolant pressure hose.
3408539	Sensor, temperature	1	Exhaust gas temperature sensor left bank 1.
3408543	Sensor, temperature	13	Exhaust gas temperature sensor all except left bank 1, right bank 4, left bank 5.
3408544	Sensor, temperature	2	Exhaust gas temperature sensor left bank 5 and right bank 4.
3408556	Sensor, temperature	3	Left bank front intake manifold (boost) pressure sensor, right bank rear intake manifold (boost) pressure sensor, and coolant pressure sensor.

3408558	Sensor, temperature	2	Pre/post lube filter pressure sensor.
3408562	Sensor, temperature	1	Fuel rail pressure sensor (part of electronic control valve assembly).
3408564	Sensor, temperature	1	Timing rail pressure sensor (part of electronic control valve assembly).
3408566	Sensor, temperature	1	Fuel pump pressure sensor.
3410392	Elbow, male union	1	Number 20 fuel filter head outlet fitting.
3627412	Clip	1	Secures harness breakout for fuel pump pressure sensor and actuator to the cam follower cover at the lower capscrew location.
3628553	Brace, tube	3	L-bracket mounted to hand hole cover left bank 3 for quantum electronic control module ground breakout.
L-bracket for securing coolant pressure sensor hose to the left bank 1 hand hole cover.			
L-bracket mounted on hand hole cover left bank 7 for Cense™ ground breakout.			
368585	Brace, tube	1	L-bracket to secures Centinal™/OEM speed sensor breakout to block at the rear left of the engine.

3629452	Clip	1	Secures right bank inlet manifold temperature sensor harness breakout to cam cover at the upper capscrew location.
3900589	Nut, hexagonal head flange	3	M8 secures quantum ground breakout to L-breakout on the left bank 3 hand hole cover.
Secures p-clip holding Centinal™ harness breakout to L-bracket on block at rear left of the engine.			
Secures Cense™ ground breakout to L-bracket on the left bank 7 hand hole cover.			
3900629	Screw, hexagonal flange capscrew	6	M8 secures quantum electronic control module ground breakout to L-bracket on the left bank 3 hand hole cover.
Secures Centinal™ harness breakout to L-bracket on block at rear left of the engine.			
Mounts p-clip onto support brace on the aftercooler covers at the right bank rear of the engine.			

Mounts p-clip onto support braces on the right bank front and left bank rear aftercooler covers towards the middle of the engine.			
Secures Cense™ electronic control module ground breakout to tube brace mounted on the left bank hand hole cover.			
3900630	Screw, hexagonal flange capscrew	1	M8 mounts p-clip onto bracket on front left aftercooler cover.
3900635	Screw, hexagonal flange head capscrew	8	Secures exhaust gas temperature sensor harness boxes to aftercooler covers.
3900679	Screw, hexagonal flange capscrew	3	M10 mounts harness box at the rear of the engine to the flywheel housing or support bracket.
Mounts the front section of the quantum harness to front cam follower cover at lower capscrew location.			
3906216	Nut, hexagonal head flange	1	M6 secures p-clip onto stud on the bracket on left bank front fuel rail.
3913372	Screw, hexagonal flange capscrew	1	M12 secures p-clip holding blowby breakout to oil pan adapter.
4007074	Elbow, male union	1	Fuel pump number 20 inlet fitting.

4066553	Bracket, electronic control module	1	Mounting bracket for Cense™ electronic control module.
4066560	Bracket, wiring retainer	1	Z-bracket mounted on rock housing below the water transfer tubes (left bank 1).
4066663	Brace, wiring	1	Secures exhaust gas temperature sensor harness to rocker housing (left bank 1 only).
4066935	Brace, wiring	2	Mounts on right bank rear aftercooler cover and supports exhaust gas temperature sensor harness breakout connecting to the main harness.
Brace to secure the fan clutch/coolant temperature harness breakout and the exhaust gas temperature sensor harness breakout on the left bank front aftercooler cover.			
4066937	Brace, wiring	2	Mounts on the right bank front and the left bank rear aftercooler cover towards the middle of the engine to support the exhaust gas temperature sensor harness breakouts connecting to the main harness.

4067050	Harness, electronic control module wiring	1	Engine harness, mounted on the left bank with Cense™ option. Harness will have Packard spade fuses, sealed OEM box, potted back shells, Y's and T's and Schlemer conduit.
4067051	Harness, electronic control module wiring	1	Right bank Cense™ harness.
4067052	Harness, electronic control module wiring	2	Right bank front and left bank rear exhaust gas temperature sensor harness boxes.
4067053	Harness, electronic control module wiring	2	Left bank front and right bank rear exhaust gas temperature sensor harness boxes.
4086486	Harness, electronic control module wiring	1	Rear section of the engine harness for Cense™ option (mounted on flywheel housing or support bracket on flywheel housing).
4088734	Sensor, pressure	1	Oil rifle pressure sensor (transparent gauge 0 to 100 psig) mounted on flywheel housing.
4097481	Brace, wiring	13	Secures exhaust gas temperature sensor harnesses to rocker housing (all except left bank 1, left bank 5, and right bank 4).

4100552	Clip	4	Secures Centinal™ harness breakout to L-bracket on the block at the rear left of the engine.
Secures fuel pump pressure sensor breakout to fuel pump flange.			
Secures coolant pressure sensor harness breakout to cam covers.			
4100627	Hose, flexible	1	Coolant pressure sensor hose adapter.
1400631	Adapter, o-ring	1	Adapter fitted into cylinder block in the kim-hot start location.
4101147	Hose, flexible	1	Number 20 fuel filter hose - fuel pump hose with 90 degree elbow.
QSK60 Engines with Single Cylinder Air Compressors Add to Kit 60-F			
4100720	Hose, flexible	1	Fuel pump - electronic control valve assembly fuel hose.
QSK60 Engines with Twin Cylinder Air Compressors Add to Kit 60-F			
4080587	Hose, flexible	1	Fuel pump - electronic control valve assembly fuel hose.
QSK60 Engines Without an Air Compressors Add to Kit 60-F			

4065606	Hose, flexible	1	Fuel pump - electronic control valve assembly fuel hose.
QSK60 Engines With "00" Flywheel Housing Add to Kit 60-F			
3408471	Sensor, speed	1	Four pin single plug speed sensor for "00" flywheel housing.
QSK60 Engines With "0" Flywheel Housing Add to Kit 60 -F			
3408472	Sensor, speed	1	Four pin single plug speed sensor for "00" flywheel housing.
3902460	Screw, hexagonal flange capscrew	2	M10 mounts bracket onto the flywheel housing.
4067011	Bracket, electronic control module wiring	1	Bracket to mount rear section of the harness box on the flywheel housing.
QSK60 Engines Without Eliminator Add to Kit 60-F			
3062329	Clamp, wire tie	1	
4066653	Harness, electronic control module wiring	1	Lubrication filter extension harness.
4100552	Clip	4	Secures the extension harness to cam covers and lubrication filter head.
QSK60 Engines With Eliminator Add to Kit 60-F			
3900626	Screw, hexagonal flange head capscrew	1	Secures p-clip to Cense™ harness box.

4066652	Harness, electronic control module wiring	1	Eliminator extension harness.
QSK60 Engines With Rockford Fan Clutch Solenoid Add to Kit 60-F			
114307	Brace, tube	1	Brace to support the fan clutch breakout extension harness.
3093956	Screw, hexagonal flange head capscrew	1	M8 secures p-clip to fan clutch breakout support brace.
3900589	Nut, hexagonal flange	1	M8 secures p-clip to fan clutch breakout support brace.
3903990	Screw, hexagonal flange head capscrew	1	M10 mounts support brace to the idler pulley bracket.
3629452	Clip	1	Secures fan clutch breakout to support brace.
4066664	Harness, electronic control module wiring	1	Fan clutch extension harness.
QSK60 Engines With Burn and Make-up Centinal™ Oil Replenishing System Add to Kit 60-F			
3170823	Elbow, male union	1	
3682300	Union, male	1	
3900635	Screw, hexagonal flange head capscrew	2	
3901249	Screw, hexagonal flange head capscrew	2	
4000694	Connector, male	1	
4066833	Bracket, valve	1	
4086294	Union, male	1	
4086355	Hose, flexible	1	

4086357	Hose, flexible	1	
4086648	Valve, oil control	1	
3903990	Screw, hexagonal flange head capscrew	3	
4086288	Adapter, o-ring	1	
4086356	Hose, flexible	1	
4086382	Clamp, retaining	1	
QSK60 Engines With Burn Only Centinal™ Oil Replenishing System Add to Kit 60- F			
3170823	Elbow, male union	1	
3682300	Union, male	1	
3900635	Screw, hexagonal flange head capscrew	2	
3901249	Screw, hexagonal flange head capscrew	2	
4000694	Connector, male	1	
4066833	Bracket, valve	1	
4086294	Union, male	1	
4086355	Hose, flexible	1	
4086357	Hose, flexible	1	
4086648	Valve, oil control	1	
4016411	Elbow, male union	1	
4016412	Adapter, reducing	1	
4016565	Hose, flexible	1	

Kit 60-J Crankcase Breather, Hand Hole Covers and Mounting Hardware for QSK60 Engines Upgrading From Head Deck Breathers to Phase 2.0 Harness and Breather			
Part Number	Name	Quantity	Remarks
4101308	Cover, camshaft	1	Cam cover at left bank.
4085924	Gasket, camshaft cover	1	Gasket for cam cover.
3900679	Screw, hexagonal flange head capscrew	5	Screws
3901448	Screw, hexagonal flange head capscrew	2	Screws
3902460	Screw, hexagonal flange head capscrew	4	Screws breather box
3075658	Seal, o-ring	1	Seal tube
4007901	Hose, plain	1	Inlet hose
43828 F	Clamp, hose	2	Hose Clamps
4086563	Hose, plain	1	Outlet hose
43828 B	Clamp, hose	2	Hose clamps
3635341	Tube, air	1	Y-tube
4086143	Hose, plain	1	Short hose to sensor
209773	Clamp, hose	4	Hose clamps
4086562	Hose, plain	1	Long hose
4086561	Hose, plain	1	From the sensor hose
114307	Brace, tube	1	Brace
68189	Clip	2	P-clips
3093828	Nut, hexagonal flange	1	Screw
3903990	Screw, hexagonal flange head capscrew	1	Nut

4086695	Cover, hand hole	3	Plain covers
4017621	Plug, expansion	1	Expansion plugs for head deck
QSK60 Engines With Rear Double Deep Oil Pan Add to Kit 60-J			
4100596	Breather, crankcase	1	Breather box
4086585	Elbow, male adapter	1	Elbow
4100638	Hose, flexible	1	Oil return hose
4100719	Hose, flexible	1	Oil return hose
4086753	Clamp, retaining	2	Clamp
3901249	Screw, hexagonal flange head capscrew	1	Screw for clamp
3903210	Nut, hexagonal flange	1	Nut
QSK60 Engines With Center Double Deep Oil Pan Add to Kit 60-J			
4101381	Breather, crankcase	1	Breather box
4101097	Elbow, male adapter	1	Elbow
4101407	Hose, flexible	1	Oil return hose
4100997	Hose, flexible	1	Oil return hose
4086753	Clamp, retaining	2	Clamp
3903990	Screw, hexagonal flange head capscrew	1	Screw clamps to blankets
3089016	Screw, hexagonal flange head capscrew	1	Screw
4101402	Bracket, breather	1	Bracket 1
4101403	Bracket, breather	1	Bracket 2
QSK60 Engines With Front Double Deep Oil Pan Add Kit 60-J			
4100596	Breather, box	1	Breather box
4100641	Elbow, male adapter	1	Elbow 90 degrees
4100637	Hose, flexible	1	Oil return hose
4100717	Hose, flexible	1	Oil return hose

4086753	Clamp, retaining	2	Clamp
3901249	Screw, hexagonal flange head capscrew	1	Screw
3903210	Nut, hexagonal flange	1	Nut
3628585	Brace, tube	1	Bracket for drain hose
3629451	Clip	1	P-clip
3902662	Nut, hexagonal flange	1	Nut
3093957	Screw, hexagonal flange head capscrew	1	Screw
QSK60 Engines With Short Dip at Position Left Bank 3 Add to Kit 60-J			
4100687	Cover, hand hole	1	Inlet hand hole cover
4100660	Seal o-ring	3	Hand hole cover o- ring
4066212	Hose, plain	1	Inlet hose
43828 F	Clamp, hose	2	Hose clamp
4100764	Bracket, oil tube	1	Bracket filler
3902460	Screw, hexagonal flange head capscrew	2	Clamp screws
4100767	Tube, breather	1	Inlet tube
131026	Seal, o-ring	1	O-ring dipstick
4080456	Dipstick	1	Dipstick
QSK60 Engines With Extended Dip at Position Left Bank 3 Add to Kit 60-J			
4100685	Cover, hand hole	1	Inlet hand hole cover
4100660	Seal, o-ring	3	Hand hole cover o- ring
4066212	Hose, plain	1	Inlet hose
43828 F	Clamp, hose	2	Hose clamp
4100764	Bracket, oil tube	1	Bracket filler

3902460	Screw, hexagonal flange head capscrew	4	Clamp screws
4100877	Bracket, oil tube	1	Dipstick brace
108722	Clip	1	P-clip
3093957	Screw, hexagonal flange head capscrew	3	Screw
3093828	Nut, hexagonal flange	1	Nut
4100767	Tube, breather	1	Inlet tube
4100883	Bracket, oil tube	1	Top dipstick tube bracket
131026	Seal, o-ring	1	O-ring dipstick
4080458	Dipstick	1	Dipstick
QSK60 Engines With No Dipstick at Position Left Bank 3 Add to Kit 60-J			
4100768	Cover, hand hole	1	Inlet hand hole cover
4100660	Seal, o-ring	3	Hand hole cover o- rings
4066212	Hose, plain	1	Inlet hose
43828 F	Clamp, hose	2	Hose clamp
Kit 60-N Crankcase Breather, Hand Hole Covers and Mounting Hardware for QSK60 Engines Upgrading From Single Bank Breathers to Phase 2.0 Harness and Breather			
Part Number	Name	Quantity	Remarks
4101308	Cover, camshaft	1	Cam cover at left bank
4085924	Gasket, Camshaft cover	1	Gasket for cam cover

3900679	Screw, hexagonal flange head capscrew	5	Screws
3901448	Screw, hexagonal flange head capscrew	2	Screws
3902460	Screw, hexagonal flange head capscrew	4	Screws breather box
3075658	Seal, o-ring	1	Seal tube
4007901	Hose, plain	1	Inlet hose
43828 F	Clamp, hose	2	Hose Clamps
4086563	Hose, plain	1	Outlet hose
43828 B	Clamp, hose	2	Hose clamps
3635341	Tube, air	1	Y-tube
4086143	Hose, plain	1	Short hose to sensor
209773	Clamp, hose	4	Hose clamps
4086562	Hose, plain	1	Long hose
4086561	Hose, plain	1	From sensor hose
114307	Brace, tube	1	Brace
68189	Clip	2	P-clips
3093828	Nut, hexagonal flange	1	Screw
3903990	Screw, hexagonal flange head capscrew	1	Nut
4086695	Cover, hand hole	2	Plain covers
QSK60 Engines With Double Deep Rear Oil Pan Add to Kit 60-N			
4100596	Breather, crankcase	1	Breather box
4086585	Elbow, male adapter	1	Elbow
4100638	Hose, flexible	1	Oil return hose
4100719	Hose, flexible	1	Oil return hose
4086753	Clamp, retaining	2	Clamp
3901249	Screw, hexagonal flange head capscrew	1	Screw for clamp
3903210	Nut, hexagonal flange	1	Nut

QSK60 Engines With Center Double Deep Oil Pan Add to Kit 60-N			
4101381	Breather, crankcase	1	Breather box
4101097	Elbow, male adapter	1	Elbow
4101407	Hose, flexible	1	Oil return hose
4100997	Hose, flexible	1	Oil return hose
4086753	Clamp, retaining	2	Clamp
3903990	Screw, hexagonal flange head capscrew	1	Screw clamps to blankets
3089016	Screw, hexagonal flange head capscrew	1	Screw
4101402	Bracket, breather	1	Bracket 1
4101403	Bracket, breather	1	Bracket 2
QSK60 Engines With Front Double Deep Oil Pan Add to Kit 60-N			
4100596	Breather, box	1	Breather box
4100641	Elbow, male adapter	1	Elbow 90 degree
4100637	Hose, flexible	1	Oil return hose
4100717	Hose, flexible	1	Oil return hose
4086753	Clamp, retaining	2	Clamp
3901249	Screw, hexagonal flange head capscrew	1	Screw
3903210	Nut, hexagonal flange	1	Nut
3628585	Brace, tube	1	Bracket for drain hose
3629451	Clip	1	P-clip
3902662	Nut, hexagonal flange	1	Nut
3093957	Screw, hexagonal flange head capscrew	1	Screw

QSK60 Engines With Short Dip at Position Left Bank 3 Add to Kit 60-N			
4100687	Cover, hand hole	1	Inlet hand hole cover
4100660	Seal, o-ring	3	Hand hole cover o-ring
4066212	Hose, plain	1	Inlet hose
43828 F	Clamp, hose	2	Hose clamp
4100764	Bracket, oil tube	1	Bracket filler
3902460	Screw, hexagonal flange head capscrew	2	Clamp screws
4100767	Tube, breather	1	Inlet tube
131026	Seal, o-ring	1	O-ring dipstick
4080456	Dipstick	1	Dipstick
QSK60 Engines With Extended Dip at Position Left Bank 3 Add to Kit 60-N			
4100685	Cover, hand hole	1	Inlet hand hole cover
4100660	Seal, o-ring	3	Hand hole cover o-ring
4066212	Hose, plain	1	Inlet hose
43828 F	Clamp, hose	2	Hose Clamp
4100764	Bracket, oil tube	1	Bracket filler
3902460	Screw, hexagonal flange head capscrew	4	Clamp screws
4100877	Bracket, oil tube	1	Dipstick brace
108722	Clip	1	P-clip
3093957	Screw, hexagonal flange head capscrew	3	Screw
3093828	Nut, hexagonal flange	1	Nut
4100767	Tube, breather	1	Inlet tube
4100883	Bracket, oil tube	1	Top dipstick tube bracket
131026	Seal, o-ring	1	O-ring dipstick

4080458	Dipstick	1	Dipstick
QSK60 Engines With No Dipstick at Position Left Bank 3 Add to Kit 60-N			
4100768	Cover, hand hole	1	Inlet hand hole cover
4100660	Seal, o-ring	3	Hand hole cover o-rings
4066212	Hose, plain	1	Inlet hose
43828 F	Clamp, hose	2	Hose clamp

Last Modified: 28-Jul-2009
